



Ranked choice voting with different elimination procedures

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Received: 6 November 2025 / Accepted: 2 June 2026
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Abstract

Ranked choice voting (RCV) is one of the fastest-growing electoral reforms in the United States, yet few scholars have considered the degree to which RCV could be improved with a different elimination procedure. This paper examines RCV with three different elimination procedures: plurality elimination, negative plurality elimination, and Borda elimination. After some analytical observations, we examine how often RCV with each of these procedures violates three criteria separately: the Condorcet winner criterion, independence of eliminated alternatives, and monotonicity using data from RCV elections in the United States as well as simulated data from the Impartial Culture condition with some trailing rankings removed. Our results show that while there are few differences in US election data, Borda elimination consistently outperforms the traditional plurality elimination by a statistically significant amount in our simulations. Nevertheless, the magnitude of those difference are often small.

Keywords Ranked choice voting · Elimination procedures · Partial ballots · US elections · Social choice

1 Introduction

Ranked Choice Voting (RCV), also known as Instant Runoff Voting, is one of the most broadly endorsed electoral reforms in the United States today. Republicans, like Mitt Romney, and Democrats, like Elizabeth Warren, have endorsed it, claiming it tempers political polarization and strengthens “the principle of majority rule while ... including those in the

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minority.”¹ RCV has become so popular that a member of the US House of Representatives introduced a bill in Congress which required RCV for the House and Senate.²

RCV is a preferential voting method that allows voters to rank any number of candidates in order of their preference. If a candidate receives more than half of the first ranked votes, that candidate wins. If no candidate receives a majority, a candidate is eliminated, and lower-ranked candidates are moved up to fill the missing ranks. The process repeats until one candidate obtains a majority among the non-eliminated candidates.³

Traditionally, the candidate with the fewest plurality votes is eliminated (PE). However, other elimination procedures could be used instead. The candidate with the most last place votes could be removed, as suggested by Coombs (1964).⁴ His idea is equivalent to negative plurality elimination (NPE) in which voters vote for every candidate except their least preferred candidate then the candidate with the least votes is eliminated. Finally, the candidate with the least Borda Count could be eliminated (BE), which Foley and Maskin (2022), and Foley (2023), call the Total Vote Runoff.⁵ They advocate this elimination procedure because it always selects a Condorcet winner when one exists.

Each elimination procedure represents a distinct implementation of RCV, yet the effects of those procedures on RCV have not been fully explored. This study evaluates how three elimination procedures, PE, NPE, and BE, affect RCV’s compliance with three normative criteria: the Condorcet winner criterion, independence of eliminated alternatives (IEA), and upward monotonicity (hereafter monotonicity). Rather than noting that one version always adheres to a criterion while others do not, as done in the axiomatic approach (e.g., Arrow 1951), we consider the degree to which one version outperforms another by analyzing the relative proportion of ballots that have a violation. We use two types of data, those from RCV elections in the United States and ballots simulated from the Impartial Culture condition with some trailing ranks removed. We also consider partial ballots from a two-dimensional spatial voting model in the electronic supplementary material.

We find that the traditional method of eliminating candidates with the fewest plurality votes is not the best way to comply with these democratic principles. Although there are few observable differences in US elections, eliminating the candidate with the least Borda Count (RCV-BE) improves RCV’s adherence to the Condorcet Winner criterion, the IEA criterion, and the monotonicity criterion in all of our simulated elections. Examining the degree of difference, partial ballots, and elections with more than three candidates is largely new. We also provide analytical results that link Condorcet winners to IEA and monotonicity independent of any preference distribution.

¹ Wilburn (2023). Quote from Warren and Raskin (2020).

² Congress.gov. “Text - H.R.9578–118th Congress (2023–2024): Ranked Choice Voting Act.” September 12, 2024. <https://www.congress.gov/bill/118th-congress/house-bill/9578/text>, Accessed October 15, 2025.

³ A voter’s ballot is exhausted (or inactive) if all the candidates on their ballot have been eliminated from the election, which is common in RCV elections because many voters do not rank all of the candidates.

⁴ The term “Coombs rule” has been also used to describe a voting rule that eliminates the candidate with the most last place votes without checking whether a candidate has a majority of first place votes first. The process continues until all candidates are eliminated. The last remaining candidate is the winner (Tideman 2006; Dougherty and Heckelman 2020). Without conducting a majority vote each round, Coombs can select differently.

⁵ The Baldwin rule eliminates the alternative with the fewest Borda points each round without evaluating whether a candidate has a majority of votes at the beginning of each elimination round. With complete ballots, RCV-BE and Baldwin select equivalently. We describe the relationship with partial ballots below.

2 The 2022 special election in Alaska and three criteria

We begin with a case study of the 2022 Alaska, Special Election for the U.S. House. See Table 1. In that year Republicans Nick Begich and Sarah Palin ran against Democrat Mary Peltola. The candidates received 53,834, 59,010, and 75,819 first place votes, respectively. Begich and Palin split the Republican vote, leaving no candidate with a majority in the field.

Using the traditional RCV-PE procedure, Peltola won the election. In the first round, Begich had the fewest plurality votes and was eliminated. In the second round, Peltola beat Palin head-to-head 91,307 to 86,175.

Interestingly, Peltola was not the Condorcet winner. A Condorcet winner is the candidate that beats every other candidate in pairwise contests. The Condorcet Winner Criterion requires a voting rule to select such a candidate if one exists (Condorcet 1785). To see its rationale, consider a sports competition such as boxing. If a boxer defeats all contenders in their weight class, that boxer ought to be crowned champion of the weight class. Similarly, if a candidate beats all other candidates head-to-head, Condorcet reasoned that the candidate ought to win the election—a notion most social choice scholars would accept. The Condorcet winner criterion makes the notion of “majority winner” precise, which allows us to evaluate how well the three procedures select candidates with majority support ... an imprecise goal of RCV proponents.⁶

In the 2022 election, the Condorcet winner was Nick Begich, who was eliminated in the first round. Begich had more head-to-head votes than Palin (101,304 to 63,709) and more head-to-head votes than Peltola (87,989 votes to 79,497). Because RCV-PE eliminated the Condorcet winner in the first round, RCV-PE violated the Condorcet Winner Criterion in this election.

RCV-PE also violated IEA. IEA, also known as the avoidance of the spoiler effect, stipulates that a candidate who wins an election should not lose the election if one or more of the non-winning candidates are removed. A system that violates IEA is vulnerable to selecting a different winner depending on who enters or leaves a race, potentially delegitimizing a democratic contest. If a system violates IEA often, political parties may try to manipulate the result by pressuring third-party candidates to drop out, or rewarding third party candidates for joining the race, thereby undermining the goal of multi-candidate elections.

If Sarah Palin was eliminated from the 2022 election, Begich would defeat Peltola, creating a different winner than Peltola even though Peltola would remain in the race. Hence, RCV-PE violates the IEA criterion in this election.

Furthermore, RCV-PE violates monotonicity. Monotonicity dictates that if a candidate is winning an election, and one or more voters rank that candidate higher on their ballot(s), while all other rankings remain the same, then that candidate should not lose the election. In essence, monotonicity tests whether the voting rule is consistent with honest voting. Most

Table 1 Ballot results from the 2022 Alaska, US house special election

	27,165	15,488	11,181	34,155	3,678	21,177	47,470	4,699	23,650
rank1	Begich	Begich	Begich	Palin	Palin	Palin	Peltola	Peltola	Peltola
rank2	Palin	Peltola		Begich	Peltola		Begich	Palin	
rank3	Peltola	Palin		Peltola	Begich		Palin	Begich	

For simplicity, write-in candidates are excluded from the table

⁶ Ranked Choice Voting, FairVote, <https://fairvote.org/our-reforms/ranked-choice-voting/>.

people assume that if they switch their vote to a winning candidate, the switch will help the candidate win, not make the candidate lose. If a voting rule violates monotonicity, voters may feel penalized for ranking a favorite candidate more honestly and demand changes to the voting rule.

To see why RCV-PE violated monotonicity in the 2022 Alaska election, notice that if 5,177 of the voters in the sixth column ranked Peltola first, rather than Palin, Palin would have 53,833 first place votes compared to Begich's 53,834 and Palin would have been eliminated. Begich would defeat Peltola in the second round. Hence, more backing for Peltola would have led to her defeat.

In contrast, RCV-NPE and RCV-BE would have selected Begich as the winner, avoiding the problems suffered by RCV-PE. Under RCV-NPE, Palin would have been eliminated in the first round because she had the most last place votes – 97,789 compared to Begich's 53,204 and Peltola's 93,678. Begich would win in the second-round against Peltola. Under RCV-BE, Palin also would have been eliminated in the first round. Giving 2 points to rank1 candidates, 1 point to rank2 candidates, 0 points to rank3 candidates, and 0.5 points to unranked candidates ($0.5 = (U-1)/2$ points, where U is the number of unranked candidates) Borda produces 211,706.5 points for Begich, 167,299.5 points for Palin, and 186,983.0 points for Peltola in the first round. Because Palin has the least Borda Count, Palin would have been eliminated. In the second round, Begich would have won the head-to-head contest.

Because RCV-NPE and RCV-BE select Begich, both adhere to the Condorcet criterion in this particular election. They also adhere to the IEA criterion because reducing the set of available candidates to any pair which includes Begich produces Begich as the winner. Finally, both RCV-NPE and RCV-BE adhere to monotonicity because moving Begich up ballots to successfully eliminate Palin in the first round (resp. Peltola) leaves Begich against Peltola (resp. Begich against Palin) in the second round. Begich wins the pairwise matchup.

The election illustrates that RCV-NPE and RCV-BE can outperform RCV-PE on the criteria we examine. More generally, Smith (1973) and Fishburn and Gehrlein (1976) have shown that Borda elimination rule is the only scoring elimination rule that will always select a Condorcet winner when one exists. This is why Foley and Maskin (2022) advocate it. If Borda Count is well defined, the relationship extends to partial ballots as well.⁷ That tells us that RCV-BE will outperform RCV-PE and RCV-NPE on the Condorcet winner criterion, but it does not tell us by how much. Nor does it tell us whether RCV-BE will consistently outperform the other elimination procedures on IEA and monotonicity.

3 Literature review

Studies of three candidate elections show that the elimination versions of plurality and negative plurality generally outperform their single-stage counterparts (Gehrlein 1993; Gehrlein and Lepelley 2001). Other studies have calculated how often plurality elimination rule, negative plurality elimination rule, and Borda elimination rule select a Condorcet winner in three-candidate elections (Gehrlein 1993; Gehrlein and Lepelley 2001, 2017). The three

⁷How Borda Count treats unranked candidates affects whether RCV-BE is Condorcet consistent. It is Condorcet consistent with the definition used here. It is not Condorcet consistent using some definitions previously implemented (e.g., McCune 2023, McCune and Wilson 2023).

rules are similar to RCV-PE, RCV-NPE, and RCV-BE, respectively, but they are not identical because they do not determine whether there is a majority winner before each elimination round. The distinction is important for RCV-NPE because RCV-NPE and negative plurality elimination rule can select differently. The other two voting rules always select the same candidate as their RCV analogs.

Grofman and Feld (2004) show that RCV-NPE is more likely to select a Condorcet winner than NPE-PE in a single-dimensional model with single-peaked preferences. Gehrlein, Lepelley, and Smaoui (2011) examine three candidate elections in which voter preferences for the candidates vary as a function of the proximity of voter preferences to being perfectly single-peaked, perfectly single-troughed, or perfectly polarized, which they call group coherence. They find that plurality elimination rule, negative plurality elimination rule, and Borda count are not always more likely to select a Condorcet winner as groups become more coherent, suggesting that social structure can have complex, non-intuitive effects on voting outcomes.

In US elections, RCV-PE almost always selects the Condorcet winner (Graham-Squire and McCune 2023; McCune et al. 2025). The two US cases regularly cited include progressive Bob Kiss's victory in the 2009 Burlington, Vermont mayoral election over the Condorcet winner Andy Montroll, and Peltola's victory in the 2022 Alaska election described earlier (Ornstein and Norman 2014; Graham-Squire and McCune 2023; McCune et al. 2025).

With regards to IEA, simulated results based on the Impartial Anonymous Culture (IAC) and Impartial Culture (IC) distributions suggest that RCV-PE is substantially less likely to violate IEA than plurality rule, and the probability of violating IEA generally increases as the number of candidates increases (McCune and Wilson 2023). Simulations from a single dimensional voting model with only one candidate eliminated at a time suggest that RCV-PE adheres to IEA in roughly 48–64% of elections, and it is less likely to adhere to IEA as the number of candidates increases (Dougherty and Edward 2011). To the best of our knowledge, RCV-PE, RCV-NPE, and RCV-BE have not been compared in terms of the IEA criterion.

In US data, IEA-PE violations have been found in four elections: the 2009 Mayoral election in Burlington; the 2021 City Council, Ward 2 election in Minneapolis, Minnesota; the 2022 House election in Alaska; and the 2022 Oakland School Director election (McCune Graham-Squire and McCune 2023; McCune and Wilson 2023). With regards to monotonicity, Borda elimination rule is better at avoiding the monotonicity paradox than plurality elimination rule or negative plurality elimination rule with IAC preferences (Smaoui et al. 2016). For large electorates, monotonicity violations occur at rates of 3.07% under the Borda elimination rule, 4.51% under the plurality elimination rule, and 5.56% under the negative plurality elimination rule (ibid, p. 6, Table 3). Furthermore, while plurality elimination rule and negative plurality elimination rule (both without majority votes) can fail monotonicity when a Condorcet winner exists, with IAC preferences Borda elimination rule is completely immune to monotonicity failures in three-candidate elections when a Condorcet winner does exist (Gehrlein and Lepelley 2017).

Mathematical analysis of RCV-PE shows that susceptibility to monotonicity failure is hardly rare. The rate of violation increases as the number of candidates increases (Quas 2004) and it increases quite substantially as three-candidate elections become more competitive (Miller 2017). Using a two-dimensional spatial voting model, Ornstein and Nor-

man (2014) show that in three candidate elections a substantial proportion of competitive RCV-PE ballot profiles are vulnerable to monotonicity failure, including upwards of 50% in closely contested profiles.

In practice, however, monotonicity failures are rare. In a database of 182 RCV elections in the US, monotonicity violations occurred in only three elections: the 2009 Burlington Mayoral; the 2021 Minneapolis Ward 2; and the 2022 Alaska House (Graham-Squire et al. 2021; Graham-Squire and McCune 2023).

Our study differs from previous studies in four important ways. First our voting rules determine whether there is a majority winner before each elimination round, which is important for the distinction between RCV-NPE and negative plurality elimination rule. Second, the literature focuses on preferences generated from the IC, IAC, and other distributions, that are complete. Our simulations focus on partial ballots. Third, we derive case-based analytic results that link Condorcet winners to IEA and monotonicity independent of any assumed preference distribution. Fourth, most of the theoretical literature is limited to three candidates. We examine cases with three or more candidates.

4 Definitions

Let C be the number of balloted candidates, R be the number of possible ranks in an election ballot, and N be the number of voters. Furthermore, let CW denote the Condorcet winner, when one exists, and W denote the candidate elected by a voting rule. In the United States, RCV elections require voters to strictly rank candidates, but they do not require ballots to be complete. For example, Minneapolis allows three ranks. A Minneapolis ballot would be complete if there were three candidates and a voter ranked all three. It would be partial if a voter ranked one of the three candidates and left two candidates unranked. It also would be partial if five candidates ran in the election but only three ranks were available. Unranked candidates are common in RCV elections in the US.

We assume voters prefer the candidates they rank higher in the ballot more than the candidates they rank lower in the ballot, they prefer ranked candidates to unranked candidates, and they are indifferent among the candidates they leave unranked.

For RCV-NPE this means that all the candidates a voter leaves unranked are tied for last place. Hence, if there are three candidates $\{x, y, z\}$ and a voter only ranks x in first place, both y and z would receive one last place vote. If another voter ranks the candidates $x > y > z$, then only z would receive a last place vote from the voter.

For RCV-BE this means that the candidate with the highest ranking receives $C-1$ points, the second highest ranking receives $C-2$ points, ..., the candidate in rank R receives $C-R$ points, and unranked candidates receive $(U-1)/2$ points, where U is the number of candidates left unranked by the voter. This treatment assures that the sum of Borda points is $C(C-1)/2$ and that RCV-BE will always select a Condorcet winner if one exists.

For any round of elimination in which a voter leaves a single balloted candidate unranked, we move that candidate into their least preferred position (if such a position is available). The practice makes ballots more complete. Balloted candidates exclude write ins.

If two or more candidates are tied for elimination in any round, we eliminate a single candidate randomly. Surprisingly, randomly eliminating a single candidate is the most common method for breaking ties in the United States.⁸

5 Analytical results

In this section we relate the existence and selection of Condorcet winners to IEA and monotonicity. Assume $C > 2$ and there are no ties. We classify outcomes into three categories: necessary satisfaction, necessary violation, and indeterminate cases.

5.1 IEA

RCV-BE always selects the Condorcet winner when one exists, while RCV-PE and RCV-NPE may fail to do so. This observation yields the following proposition.

Proposition 1 (IEA and Condorcet). For any profile without ties:

- 1) If $W = CW$, then RCV-BE necessarily satisfies IEA. RCV-PE and RCV-NPE necessarily satisfy IEA for $C = 3$ but are indeterminate for $C > 3$.
- 2) If CW exists but $W \neq CW$, then RCV-PE and RCV-NPE necessarily violate IEA. Such a case cannot arise under RCV-BE.
- 3) If $\sim CW$, then all three rules necessarily violate IEA.

Reasoning. If $W = CW$, then RCV-BE selects the Condorcet winner on every relevant subset, necessarily satisfying IEA. RCV-PE and RCV-NPE necessarily satisfy IEA for $C = 3$ because W wins among all three alternatives (as it originally did) and in any subset $\{W, x\}$. For $C > 3$, the violation of IEA by RCV-PE and RCV-NPE is indeterminate. If CW exists but $W \neq CW$, the two-candidate subset $\{W, CW\}$ must select CW , so any rule choosing $W \neq CW$ from the full set violates IEA. If $\sim CW$, some candidate x must defeat W head-to-head, necessarily violating IEA with the subset $\{W, x\}$.

5.2 Monotonicity

Monotonicity is less amenable to a single proposition because outcomes depend on how movements of W change the elimination order. Two mechanisms create violations: a “majority case,” in which another candidate secures a majority before W is eliminated, and

⁸ Ties are broken randomly in Alaska; Portland, Oregon; Maine; the District of Columbia; San Francisco; Oakland, San Leandro, and Berkeley, California; Minneapolis, Minnesota; and Ft. Collins, Colorado; to name a few. Two additional methods for breaking ties include eliminating all tied candidates at once (as applied in Boulder, Colorado) and the backward tie-breaking method (as applied in Northampton and Easthampton, Massachusetts; Tacoma Park, Maryland; and Santa Fe, New Mexico). Under the backward tie-breaking method, if two or more candidates are tied, the candidate with the fewest votes in the previous round is eliminated. If the tie persists, the candidate with the fewest votes in the round before that is eliminated. The process continues until the tie is broken. If it cannot be resolved in this manner, a candidate is eliminated randomly.

an “elimination case,” in which (for $C > 3$) W is eliminated in a later round after being raised on some ballots in an earlier round.

Case 1 $W = CW$. The “majority case” is ruled out because nothing beats W head-to-head. For RCV-BE, the elimination case is also ruled out because the Condorcet winner must have an above-average Borda score on every subset and can be never eliminated. RCV-PE and RCV-NPE necessarily satisfy monotonicity for $C = 3$ because the “elimination case” cannot arise. For $C > 3$, RCV-PE and RCV-NPE violate monotonicity if and only if an elimination case can be induced (see the electronic supplementary material for an example).

Case 2 A CW exists but $W \neq CW$. This case cannot arise under RCV-BE. For RCV-PE and RCV-NPE, monotonicity is indeterminate. Violations are possible, but not necessary, because for some profiles all possible movements of W produce the same winner (see the electronic supplementary material for an example).

Case 3 $\sim CW$. For all three rules, monotonicity is indeterminate. In the absence of a Condorcet winner, neither the majority case nor the elimination case can be excluded (see the electronic supplementary material for an example of satisfaction).

5.3 Summary

Proposition 1 does not tell us the relative number of ballot profiles in which an IEA violation occurs. However, it does tell us that for any fixed set of elections, the proportion of elections in which RCV-BE satisfies IEA will be at least as large as the corresponding proportion for RCV-PE (resp., RCV-NPE), and it is strictly larger if RCV-PE (resp., RCV-NPE) fails to select the Condorcet winner in at least one election. Even though RCV-BE is at least as likely to satisfy IEA as RCV-BE and RCV-NPE, we do not know by how much.

For monotonicity, only some cases yield definitive results; others depend on the profile and whether an elimination path can be induced. Simulations help quantify the frequency of those indeterminate cases and the degree to which RCV-BE outperforms RCV-PE and RCV-NPE.

6 Data

We use two types of data in the remainder of the paper: RCV elections from the United States (2004 to 2025) and simulated data from the impartial culture condition with partial ballots.

6.1 U. S. election data

Fair Vote provides ballot data on almost every single-winner RCV election in the United States (Otis 2025). We examine all 362 of the elections with at least three candidates, 2004–2025. They include local, state, and federal elections conducted in various jurisdictions in Alaska, California, Colorado, Massachusetts, Maine, Maryland, Minnesota, New Mexico, New York, Oregon, Utah, and Vermont. The number of balloted candidates range from 138

elections with 3 candidates to the single Minneapolis Mayoral Election of 2013 with 35 candidates. Of the 362 elections examined, 115 elections have at least 5 possible ranks. Excluding write-ins, 39 have five candidates on the ballot and 8 have nine.⁹ The average election contains roughly 44,000 valid ballots, 3.9 possible ranks, and 5.2 candidates.

Voters make three common mistakes when they rank candidates. They overrank, over-vote, and skip ranks. For each, we apply the most common treatment in the United States.

An overrank occurs when a voter ranks a single candidate multiple times. For example, they select Anne as both their first choice and second choice. In such cases, we accept the highest ranking, then ignore each lower rankings for the same candidate and move the remaining candidates up the voter's ballot.¹⁰

An overvote occurs when a voter casts more than one vote for the same rank. For example, when they mark both Anne and Bob as their first choice. In these cases, we treat the ballot as exhausted as soon as an overvote is reached. Ranks above the overvote are included in the ballot profile, but overvoted candidates and candidates ranked beneath the overvote are dropped from the ballot.¹¹

A skip occurs when a voter leaves a rank blank, i.e., they skip one or more rankings. In such cases, we ignore the blank and elevate the voter's remaining rankings into the missing ranks.¹² The only time that an unranked candidate becomes ranked are cases in which a ranked position is blank and a single balloted candidate is unranked.

After processing, the US data have three noticeable characteristics. First, ballot profiles are typically skewed toward one (or two) candidates. For example, with three candidates, roughly 53% of the elections have a majority winner in the first round, with an average vote share of 59%. With five candidates, roughly 39% of the elections have a first-round majority winner with an average vote share of 61%. With nine candidates, roughly 36% of the voters rank the same candidate first.

Second, there are many partial ballots in actual RCV elections – that is rankings with at least one missing value. After processing, all blanks trail the list of ranked candidates. In elections with three candidates and three ranks, 0.156 of the ballots had a single trailing blank and 0.302 of the ballots had two trailing blanks. In the elections with five candidates and five ranks there were 0.263, 0.102, 0.201, and 0.384 trailing blanks for one missing, two missing, three missing, and four missing, respectively. We use this information to create partial rankings for our simulated data.

Third, candidates with the most votes are disproportionately the candidate with the most trailing blanks. If a voter bullet votes, meaning they rank only one candidate, they often rank the plurality winner. For example, in three candidate elections, a little less than a third of

⁹ We treat write-in candidates as a single person without affecting the results.

¹⁰ Another method for addressing overranks is to exhaust the ballot upon over ranking. In other words, election officials would treat the ballot as “exhausted” or invalid from the point of an over rank onward, as done in Telluride Colorado, https://www.amherstma.gov/DocumentCenter/View/49109/TellurideCO_RCV_Feedback_2016.

¹¹ Another approach to overvotes is to skip the overvotes and continue counting lower ranked candidates (as done in Burlington, Vermont; Minneapolis, Minnesota; and Portland, Oregon). Surprisingly, exhausting the ballot upon overvote is much more common.

¹² There are at least two other approaches to skipped ranks. First, a ballot continues after a single skipped ranking, but the ballot becomes exhausted after two or more consecutive skips (as done in Alaska and Maine). Second, the ballot is treated as exhausted when the first skipped rank is encountered (as done in Salt Lake City, Utah and Boulder, Colorado).

the ballots cast for the plurality winner contain only one ranked candidate. In five candidate elections, roughly 41% of the ballots cast for the plurality winner contained only one ranked candidate. In nine candidate elections the number was roughly 36%.

6.2 IC-partial distribution

To compare the performance of the three elimination procedures in highly competitive elections, we also examined 10,000 simulated elections derived from a variant of the Impartial Culture distribution (IC) with $N=10,000$, and $C \in \{3, 5, 9\}$. IC assumes each *permutation* of the $C!$ strict preference orders are equally likely. For our simulated data, we assume $R=C$, generate an IC distribution for N voters, then replace the last r ranks with blanks, where r is the proportion of missing ranks in the US data for that $R=C$. For example, in the U.S. elections the average number of single trailing blanks for $R=C=3$ were 0.156. For two trailing ranks they were 0.302. Hence, we eliminated the last ranked candidate in the first 1,560 of the 10,000 rows of the IC data and the last two ranked candidates for the next 3,020 rows. The other rows remained complete.¹³

While actual election data tells us the frequency of normative violations in actual elections, IC-partial data help us see differences in highly-competitive elections. An analogy may help explain why studying the idealized setting is useful. If a defective bolt on a class of airplane could make it crash in one out of 10,000 flights, engineers would not say that the bolt is fine because it rarely crashes a plane. Instead, they would home in on cases in which the bolt is likely to falter and try to find the best bolt under those circumstances. Similarly, we don't want to say that the current configuration of RCV is fine because there are small differences in US election data, where preferences are skewed and different voting rules of any kind rarely matter. Instead, we want to home in on cases where voting rules do matter, such as homogenous preferences, so we can see differences between variants of RCV. Such results may not be common, but they help improve RCV for competitive races.

7 Computer script

To determine the frequency of violations under each elimination procedure, we wrote an R script that opens ballots for a single election, calculates the winner using each elimination procedure, and determines whether the selection leads to a violation of any of the three criteria separately. We repeat the process for the 362 actual elections (resp., the 10,000 simulated elections) and report the frequency of violations in each category.

The script identifies Condorcet winner violations in the usual fashion.

7.1 IEA

To identify IEA violations each subset S of the $C-1$ candidates who lost the race are combined with the winner W . If a candidate other than W wins in any subset, the elimination procedure is marked as violating IEA for that election. An elimination procedure satisfies IEA for an election only if it selects W for all combinations of losing candidates removed.

¹³ There were few elections with 9 candidates and 9 ranks, so the proportions applied to $C=9$ had to be calculated more elaborately using elections with the same number of ranks for $C \in [7, 10]$.

7.2 Monotonicity

To identify monotonicity violations our script first determines whether a tie arises during the elimination process that produces different winners. We report such cases as instances of “tie instability” separately.¹⁴

If the resolution of a tie does not produce different winners or there is no tie in the data, we test whether the winner of the initial election W could be forced to lose if W is moved up some ballots, while other rankings remain fixed.

The key to our approach is recognizing that for the “majority case” any candidate capable of defeating W must defeat W head-to-head in the original profile. For the “elimination case,” W must successfully move prior to round $C - 3$. Denote the full set of candidates c and the set of candidates that beat W head-to-head F . To avoid examining all possible elimination orders, whose number increases factorially ($O(C!)$) as the number of candidates increases, we generate all subsets $S \subseteq c$, such that $W \in S$. If $F \neq \emptyset$, we also require $S \cap F \neq \emptyset$. Among these subsets, we identify what we call G-subsets, that is the subsets in which a candidate other than W emerges as the winner. For each G-subset, we construct compatible elimination orders by concatenating all permutations of the complement set $c \setminus S$ in front of the elements of S . Candidates outside S must be eliminated before the G-subset is reached. This subset-based search ($O(C^2)$) drastically reduces the computational burden of the procedure.

After the set of viable elimination orders have been generated, each elimination order is evaluated round by round. In each round, we first check if the targeted candidate is eliminated naturally, we then attempt to move W up in the fewest possible ballots to assure the elimination of the targeted candidate. The process continues through the elimination order until the first element of the G-subset is reached, at which point the modified profile is checked for whether it in fact produces a winner other than W . If it does, the election is classified as violating monotonicity. Otherwise, the search proceeds to the next elimination order.¹⁵

8 Simulation results

This section presents a separate table of results for each criterion. The top section of each table shows results from US elections, with combined results for all C in the left-most column and results broken out for $C=3, 5, 9$ in the remaining columns. The bottom section presents results from the IC-partial distribution for $C=3, 5, 9$.

¹⁴ Less than 2% of our data contain such a tie.

¹⁵ Graham-Squire and Zayatz (2021) approach the problem differently. Following an insight provided by Miller (2017), their script runs RCV-PE on the ballot profile to determine W and the four candidates that reach the final rounds. The script then targets a monotonicity violation by moving W up only after the field is reduced to four candidates (or three if $C=3$). The success of their algorithm depends on an assumption that any ballot modification in an earlier round leads to a detectable violation in a round with four or fewer candidates. Our script does not require this assumption. It attempts to move W every round that a targeted candidate cannot be eliminated naturally, except for elections with $C>10$. For computability reasons, our script runs the relevant elimination rule until ten or fewer candidates remain. It then applies the monotonicity search to the reduce profile. Interestingly, for RCV-PE we find the same monotonicity violations in US elections as already noted in the literature. We find new violations for RCV-NPE and RCV-BE because they have not been sought by others.

Table 2 Condorcet winner violations

Voting rule	$C \in [3, 35]$	$C = 3$	$C = 5$	$C = 9$
<i>US Elections</i>				
RCV-PE	0.006	0.007	0.026	0.000
RCV-NPE	0.003	0.000	0.000	0.000
RCV-BE	0.000	0.000	0.000	0.000
# CW	359			
<i>IC Partial</i>				
RCV-PE		0.027	0.056	0.123
RCV-NPE		0.028	0.053	0.086
RCV-BE		0.000	0.000	0.000
# CW		9239	8298	6708

The figures reported are the frequency of Condorcet winner violations given the existence of a Condorcet winner and the ballot type indicated in the section. #CW indicates the number of Condorcet winners. For the 362 US Elections, $N \in [112, 942,031]$. For the 10,000 IC-partial elections, $N=10,000$

8.1 Condorcet winner

The rate at which each of the three elimination procedures violates the Condorcet winner criterion is presented in Table 2.

Because the distribution of RCV preferences in US elections are skewed, all but three of the US elections have a Condorcet winner (2021 Minneapolis Ward 2; 2022 Oakland School Board, District 4; and 2021 Portland, ME City Council, at large). This is indicated by #CW=359 at the bottom of the top section. Interestingly, RCV almost always selects the Condorcet winner, when one exists, regardless of the elimination procedure. RCV-PE only fails to select a Condorcet winner twice (2009 Burlington Mayoral; and 2022 Alaska House) out of the 359 elections where a Condorcet winner exists, both of which correspond to what is found in the literature. RCV-NPE has one violation (2008 election for Assessor/Treasurer in Pierce County, Washington), and RCV-BE has zero violations as one would expect. The difference in violation rates are not statistically significant. Part of the reason that all three elimination procedures are so successful is that many RCV elections in the US have a majority winner in the first round.

In contrast, the proportion of Condorcet winners in the IC-partial simulations range from 0.671 for $C=9$ to 0.924 for $C=3$. RCV-BE never fails to select a Condorcet winner, as expected, but RCV-PE fails to select a Condorcet winner at a low rate as well. Even though the magnitude of the differences are small particularly for $C \leq 5$, the differences between RCV-PE and RCV-BE (and RCV-BE and RCV-NPE) are significant at the 0.001 level. RCV-NPE appears to outperform RCV-PE, but the only difference that is significant is for $C=9$. One might conclude that RCV-BE performs at least as well as RCV-PE in terms of the procedure's ability to select Condorcet winners, but when differences are noticeable, they are small.

8.2 Independence of eliminated alternatives

As shown in the top of Table 3, RCV-PE violates IEA in four elections (2009 Burlington Mayoral; 2021 Minneapolis Ward 2; 2022 Alaska House; and 2022 Oakland School Board District 4). All four have been noted in the literature (McCune 2023; Graham-Squire and

Table 3 IEA violations

	Voting rule	$C \in [3, 35]$	$C = 3$	$C = 5$	$C = 9$
The figures reported are the frequency of IEA violations given the ballot type indicated in the section. For the 362 US Elections, $N \in [112, 942, 031]$. For the 10,000 IC-partial elections, $N=10,000$	<i>US elections</i>				
	RCV-PE	0.011	0.015	0.051	0.000
	RCV-NPE	0.011	0.007	0.026	0.000
	RCV-BE	0.008	0.007	0.026	0.000
	<i>IC partial</i>				
	RCV-PE		0.101	0.242	0.506
	RCV-NPE		0.102	0.229	0.433
	RCV-BE		0.076	0.170	0.329

McCune 2023). RCV-NPE also violates IEA in four elections (2008 Pierce County; 2021 Portland City Council; 2021 Minneapolis Ward 2; and 2022 Oakland School Board), which differ from RCV-PE. RCV-BE violates IEA in three elections (2021 Minneapolis Ward 2; 2021 Portland City Council; and 2022 Oakland School Board). Of course, the RCV-NPE and RCV-BE results are new.

The small rate of violation makes sense for two reasons. First, there are candidates with outright majorities in many of these elections that are difficult to displace when other candidates are eliminated. Second, because ballots are partial, removing combinations of unranked candidates is less likely to displace the winner.

The IC-partial results produce greater rates of violation because the data are less skewed. Consistent with Proposition 1, RCV-BE has a lower rate of violation than the other two procedures for each C considered. The differences between RCV-BE and RCV-PE (resp., RCV-NPE) are significant at the 0.001 level. RCV-BE performs at least as well as the other two elimination procedures on the IEA criterion. Some of the differences are sizable, particularly for large C .

8.3 Monotonicity

The rate at which RCV violates monotonicity with each of the three elimination procedures is presented in Table 4. In the US data, RCV-PE violates monotonicity in four elections, two of which have three candidates (2022 Alaska House; and 2022 Oakland School Board) and two of which have five candidates (2009 Burlington Mayoral; and 2021 Minneapolis Ward 2), none of which are new (McCune 2023; Graham-Squire and McCune 2023). RCV-NPE violates monotonicity in the Minneapolis and Oakland elections, as well as the 2008 Pierce County. The latter election had six candidates. RCV-BE violates monotonicity in Oakland and Minneapolis. The RCV-NPE and RCV-BE results are new.¹⁶

Ironically, *all* of the elections in which a candidate *could* beat W head-to-head under an elimination procedure produced a monotonicity violation, suggesting that a lack of pairwise challengers explains the result.

Greater differences between the elimination procedures can be seen in the IC-partial data, partly because the number of elections with at least one candidate who can beat W head-to-head is greater in the IC-partial data than in the US election data—roughly 10% for the $C=3$

¹⁶ Based on the data posted on dataverse in June 2025, the 2021 Portland, ME City Council election (at large) was the only election that exhibited tie instability for RCV-PE, RCV-NPE, and RCV-BE. After a controversial recount, the tie no longer existed and Rodriguez was declared the winner <https://content.civicplus.com/ap/i/assets/flcce740-3243-4e59-ae98-0c952f56db09?cache=1800%3E>.

Table 4 Monotonicity violations

Voting rule	$C \in [3, 35]$	$C = 3$	$C = 5$	$C = 9$
<i>US elections</i>				
RCV-PE	0.011	0.014	0.051	0.000
RCV-NPE	0.008	0.007	0.026	0.000
RCV-BE	0.006	0.007	0.026	0.000
<i>IC partial</i>				
RCV-PE		0.092	0.228	0.475
RCV-NPE		0.093	0.211	0.407
RCV-BE		0.067	0.159	0.316

The figures reported are the frequency of monotonicity violations given the ballot type indicated in the section. For the 362 US Elections, $N \in [112, 942, 031]$. For the 10,000 IC-partial elections, $N=10,000$

simulated data, 40% for $C=9$ simulated data, and 1% for the US data. Hence, there is greater opportunity for “majority case” violations in the simulated data.

With IC-partial ballots, RCV-BE has the lowest rate of violation regardless of the number of candidates. The rate statistically differs from RCV-PE, and RCV-NPE, in all three cases at the 0.001 level. We conclude that RCV-BE performs at least as well as RCV-PE on the monotonicity criterion. Differences are unnoticeable to moderate depending on the type of data and number of candidates.

8.4 Other distributions

To ensure that the simulated results are not unique to IC-partial distribution, we ran three additional simulations based on a two-dimensional spatial voting model with 10,000 voters. In each simulation, preferences are single peaked, symmetric, and ordered based on closest Euclidean distance to a voter’s ideal point. Ideal points are clustered around one of three bivariate-normal distributions in the proportions 40%, 40%, and 20%, respectively. Two major coalitions have means $\mu_1 = (0.25, 0.25)$ and $\mu_2 = (0.75, 0.75)$, while a third minor coalition has mean μ_3 , which varies across the three simulations with values $\mu_3 = (0.50, 0.50)$, $(0.25, 0.75)$, and $(0.00, 1.00)$, respectively. C candidates are drawn from a single distribution with mean $\mu_c = (0.50, 0.50)$. The standard deviation for each distribution is $\sigma = (0.20, 0.20)$.

The model reflects two dominant coalitions of roughly equal size and a minor coalition in one of three positions: (1) between the two major coalitions, (2) the same distance between the two dominant coalitions in an orthogonal direction, and (3) half the distance between those locations. Complete rankings are made partial using the procedure described previously.

The results from these simulations are reported in the electronic supplementary material. Consistent with our IC-partial results, RCV-BE outperforms RCV-PE on all three criteria at statistically significant rates with differences in performance larger than the IC-partial results in some cases and smaller in others.

9 Conclusion

The normative performance of Ranked Choice Voting (RCV) is not only defined by majority voting itself but also by the choice of its elimination procedure. By systematically comparing the traditional RCV-PE to two variants, RCV-NPE and RCV-BE, we have shown that

RCV-BE reduces violations of the Condorcet Winner Criterion, Independence of Eliminated Alternatives, and Monotonicity.

Prior to our study it was widely known that RCV-BE always selects a Condorcet winner when one exists. We extend the analysis to partial ballots and show the degree to which RCV-BE outperforms RCV-PE. In US elections, the difference is negligible. In our simulated data, the difference is small but significant.

In a set of election without ties, RCV-BE will satisfy IEA at least as often as RCV-PE and RCV-NPE – more often if the latter fail to select a Condorcet winner in at least one election. In US elections, the difference in performance between RCV-BE and the other two rules on the IEA criterion is negligible. In our simulated elections, RCV-BE can outperform RCV-PE by as much as 18% points in the IC-partial data and as much as 41% points in the two-dimensional data. Differences increase as C increases.

Prior to our study it was also known that with IAC preferences RCV-BE will satisfy monotonicity if a Condorcet winner exists and $C=3$. It is easy to show that this extends to other preference distributions with more candidates and partial ballots. Furthermore, if RCV-PE and RCV-NPE select the Condorcet winner, they too are immune to monotonicity violations for $C=3$ but they are not immune for $C>3$. In US election data, the difference in performance on the monotonicity criterion is negligible. In our IC-partial data, RCV-BE can outperform RCV-PE by as much as 7% points – a worthy consideration if RCV is to be applied to competitive elections.

Jurisdictions that want to improve RCV should consider two factors. First, RCV is more likely to satisfy three widely accepted criteria if Borda elimination is used in place of plurality elimination. In stylized contexts, such as those captured by the IC-partial distribution, improvement will be statistically significant. Second, whether such improvements are large enough to justify the change is a matter of interpretation. The value of the improvement has to offset the cost of making the change and the ability to convince voters that the change is important.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11127-026-01427-0>.

Acknowledgements We would like to thank the associate editor, Marek Kaminski, and two anonymous referees for useful feedback.

Author contributions Katzman and Dougherty both wrote code, conducted literature review/background studies, and wrote the manuscript.

Funding The authors did not receive support from any organization for the submitted work.

Data availability Data generated from the IC-partial distributions and two-dimensional models are publicly available upon request <dougherk@uga.edu>.

Declarations

Conflict of interest The authors declare no competing interests.

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